

Capstone Project and Test Summary

Demo Web Shop Automation Testing using Playwright JavaScript

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Field	Details
Student Name	Ajay Bhatnagar
Training Program	Wipro SDET Playwright Capstone
Project Type	Web application automation testing
Application Under Test	Demo Web Shop
Website URL	https://demowebshop.tricentis.com
GitHub Repository	https://github.com/AjayBhatnagar1712/Wipro_SDET_Playwrite_Capstone
Hosted Allure Report	https://ajaybhatnagar1712.github.io/Wipro_SDET_Playwrite_Capstone/
Automation Stack	Playwright Test, JavaScript, Node.js, Allure Report, GitHub Actions, GitHub Pages

Project Summary

The project validates a public e-commerce demo application through 6 core modules and 2 add on modules. Each module contains 15 functional test cases, giving 120 functional test cases and 360 browser executions across Chromium, Firefox, and WebKit.

Overview

The Application Under Test is Demo Web Shop, a public web-based e-commerce demo platform. It contains common online shopping features such as authentication pages, product categories, search, cart, wishlist, account-related pages, shipping information, checkout steps, and customer support pages.

This report explains the testing work completed for the Playwright JavaScript capstone project. The testing approach focuses on stable customer-facing workflows that can run locally and through GitHub Actions without depending on private user data or manual verification.

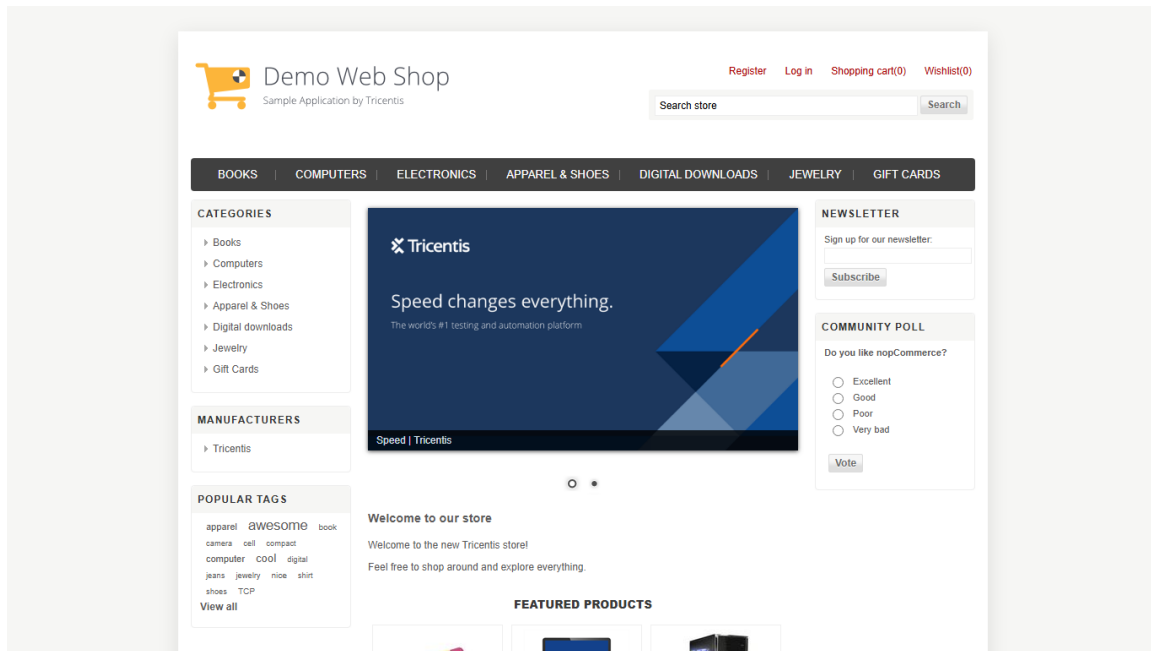


Figure 1: Demo Web Shop home page used as the main entry point.

Goals

1 Business Objectives

- Validate important e-commerce user journeys such as account access, product browsing, cart, wishlist, shipping information, checkout-related checks, and customer support information.
- Create reliable automated regression coverage that can be reviewed by the trainer through a hosted Allure report.
- Maintain a professional project structure that looks like a real Playwright engineering project.

2 Quality Engineering Objectives

- Design exactly 15 test cases per module and cover 8 modules in total.
- Use Playwright assertions to verify page visibility, headings, labels, controls, redirects, and product information.
- Run the same functional test suite on Chromium, Firefox, and WebKit to prove cross-browser coverage.

- Publish automation evidence using Allure Report and GitHub Pages.

Specifications

The capstone scope is organized into 6 core modules and 2 add on modules. Each module contains 15 functional test cases.

Module Type	Module Name	Test Cases	Requirement
Core Module	Authentication	15	FR-01
Core Module	Product and Search	15	FR-02
Core Module	Cart	15	FR-03
Core Module	Wishlist and Compare	15	FR-04
Core Module	User Profile and Account	15	FR-05
Core Module	Address and Shipping	15	FR-06
Add on Module	Add on - Checkout and Payment	15	FR-07
Add on Module	Add on - Customer Support and Information	15	FR-08

In-Scope Functional Modules

- Authentication: Validate login, registration, password recovery, protected redirects, and guest authentication links.
- Product and Search: Validate catalog navigation, product listing pages, product details, sorting controls, and product search.
- Cart: Validate cart access, empty cart messaging, add-to-cart controls, checkout progress labels, and cart-related links.
- Wishlist and Compare: Validate wishlist, compare products, recently viewed products, email-a-friend entry points, and supporting links.
- User Profile and Account: Validate account access rules for guest users, registration form fields, account footer links, and account recovery paths.
- Address and Shipping: Validate shipping and returns content, address redirects, product availability text, and checkout address/shipping labels.
- Add on - Checkout and Payment: Add coverage for checkout progress, empty-cart checkout behavior, gift card fields, add-to-cart availability, and price information.
- Add on - Customer Support and Information: Add coverage for contact us, sitemap, privacy, conditions, about, news, blog, newsletter, social links, and footer information.

Out-of-Scope Areas

- Real payment processing is not performed because Demo Web Shop is a public demo website.
- Private database validation is not included because the website is not created or hosted by us.
- CAPTCHA or Cloudflare-based flows are not part of this project because the selected website should remain stable for automated testing.

Milestone 1 - Analysis and Planning

1.1 Introduction

The analysis phase identifies what needs to be tested, how the test cases will be grouped, and how the final evidence will be shown. Since this is an automation testing capstone, the main focus is test planning, traceability, automation design, reporting, and CI/CD execution.

1.2 Requirement Analysis

Functional requirements define what the application should do. Non-functional requirements define quality expectations such as reliability, browser support, maintainability, traceability, and reporting.

1.2.1 Functional Requirements (FR)

Req ID	Module	Requirement Description	Priority
FR-01	Authentication	The application should expose login, registration, password recovery, and protected account access behavior.	High
FR-02	Product and Search	The catalog should allow users to browse categories, view product details, and search product keywords.	High
FR-03	Cart	The shopping cart should open correctly, display empty cart information, and expose cart-related controls and checkout steps.	High
FR-04	Wishlist and Compare	The site should expose wishlist, compare products, recently viewed products, and email-a-friend journeys.	Medium
FR-05	User Profile and Account	Account, order history, address book, registration, login, and recovery pages should behave correctly for a guest user.	High
FR-06	Address and Shipping	Shipping returns, product availability, delivery text, and address/shipping journey labels should be visible and correct.	Medium
FR-07	Add on - Checkout and Payment	Checkout progress, empty-cart checkout guard, payment labels, gift card inputs, and price displays should be verified.	High
FR-08	Add on - Customer Support	Customer service pages, information footer links, newsletter	Medium

	and Information	controls, and social links should be available.	
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1.2.2 Non-Functional Requirements (NFR)

Req ID	Quality Area	Requirement Description	Priority
NFR-01	Browser Compatibility	The same functional suite should run through Playwright projects for Chromium, Firefox, and WebKit.	High
NFR-02	Reliability	Tests should use stable guest-user flows and visible UI assertions to avoid dependency on temporary user data.	High
NFR-03	Maintainability	Page Object Model, shared fixtures, and reusable data should keep test updates centralized and readable.	High
NFR-04	Traceability	Each module should have clear IDs, requirement mapping, and a documented test case list.	High
NFR-05	Reporting	Execution output should be available in Allure report format and published through GitHub Pages.	High
NFR-06	CI/CD Readiness	The project should be executable from GitHub Actions so the trainer can review the latest hosted report.	High
NFR-07	Evidence	Screenshots and report artifacts should support viva discussion and result review.	Medium

1.3 Test Strategy

A hybrid quality strategy is followed. Manual observation is used to understand the website and confirm stable flows. Automated Playwright tests are then created for repeatable regression checks. The suite is grouped module-wise so that coverage can be explained clearly during viva.

1.3.1 Manual Testing Scope

- Explore the Demo Web Shop pages and decide which flows are stable enough for automation.
- Check labels, page headings, forms, links, product pages, and footer navigation before writing automated cases.
- Avoid unstable flows that can fail because of third-party verification or changing user data.

1.3.2 Automation Testing Scope

- Automate page visibility and functional UI checks for all selected modules.
- Use Page Object Model so each module has a matching reusable page class.
- Use Allure reporting to make test results easy to review.
- Run all tests on Chromium, Firefox, and WebKit through Playwright projects.

1.4 Requirements Traceability Matrix (RTM)

The RTM links each requirement to the related module and test ID range. This helps prove that no planned module is left uncovered.

Requirement	Module	Test ID Range	Coverage Status
FR-01	Authentication	AUTH-001 to AUTH-015	Covered
FR-02	Product and Search	PROD-001 to PROD-015	Covered
FR-03	Cart	CART-001 to CART-015	Covered
FR-04	Wishlist and Compare	WISH-001 to WISH-015	Covered
FR-05	User Profile and Account	USER-001 to USER-015	Covered
FR-06	Address and Shipping	ADDR-001 to ADDR-015	Covered
FR-07	Add on - Checkout and Payment	Add on - PAY-001 to Add on - PAY-015	Covered
FR-08	Add on - Customer Support and Information	Add on - INFO-001 to Add on - INFO-015	Covered

Milestone 2 - Project Setup and Design

2.1 Tools and Technology

Area	Tool or Concept	Purpose
Automation	Playwright Test	Runs browser automation tests and provides assertions.
Language	JavaScript	Used to write readable test files and page object classes.
Runtime	Node.js	Executes the Playwright project and manages dependencies.
Design Pattern	Page Object Model	Keeps selectors and page actions separate from test files.
Reporting	Allure Report	Generates a trainer-friendly HTML execution report.
CI/CD	GitHub Actions	Runs the project automatically after code is pushed.
Hosting	GitHub Pages	Publishes the Allure report online for trainer review.

2.2 Project Folder Structure

The project follows a professional Playwright structure. Test cases are grouped by module, while shared actions are placed in page objects and reusable data files.

Folder/File	Purpose
tests	Contains module-wise Playwright spec files. Each module has 15 test cases.
pages	Contains page object classes for authentication, product, cart, wishlist, account, shipping, checkout, and support pages.
fixtures	Contains reusable Playwright fixture setup for test consistency.
data	Stores project metadata such as module coverage, browser list, and expected result count.
api	Keeps internal helper code that can be extended if API validation is added later.
docs	Contains testing plan, Playwright notes, and final project report.
.github/workflows	Contains CI/CD workflow configuration used for report generation and GitHub Pages deployment.
playwright.config.js	Stores Playwright test directory, browser projects, reporter settings, timeouts, traces, screenshots, and video settings.

2.3 Environment Information

Field	Details
Automation Tool	Playwright Test
Programming Language	JavaScript
Runtime	Node.js
Reporting	Allure Report with GitHub Pages hosting
Browsers	Chromium, Firefox, WebKit
Execution Mode	Local execution and GitHub Actions CI execution
Website Under Test	https://demowebshop.tricentis.com
Repository	https://github.com/AjayBhatnagar1712/Wipro_SDET_Playwrite_Capstone

2.4 Testing Theory Used in the Project

Page Object Model

Page Object Model is used to separate test intention from page implementation. The test file says what scenario is being validated, and the page object stores how the page is opened, located, and asserted.

Assertions

Assertions are the checkpoints of automation testing. In this project they confirm whether a page, link, heading, input field, button, product card, redirect, or text value is visible and correct.

Fixtures

Fixtures help keep repeated setup reusable. They make the test suite cleaner because shared browser setup and helper access can be prepared once and used across many test files.

Cross-Browser Testing

Cross-browser testing confirms that the same feature behaves correctly in different browser engines. Playwright supports Chromium, Firefox, and WebKit, so one functional test case is executed on all configured browser projects.

Allure Reporting

Allure Report converts test execution output into a readable web report. It shows total test count, pass/fail status, suites, timings, and detailed test history, which makes it suitable for trainer review.

Milestone 3 - Playwright Automation Testing

3.1 Test Design Methodology

The test cases are designed around expected outcomes. Each test checks one clear behavior and maps to a module. This keeps the report easy to understand and keeps debugging simpler when a failure occurs.

- Positive UI verification checks confirm that expected pages and controls are visible.
- Guest-user redirect checks confirm that protected account pages require login.
- Navigation checks confirm that important header, footer, category, and information links are available.
- Product checks confirm details such as price, availability, search result content, and add-to-cart related controls.

3.2 Module-wise Automation Design

3.2.1 Authentication

Field	Details
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Module Type	Core Module
Requirement ID	FR-01
Test Folder	tests/authentication
Page Object	pages/authentication.page.js
Test Case Count	15
Objective	Validate login, registration, password recovery, protected redirects, and guest authentication links.

3.2.2 Product and Search

Field	Details
Module Type	Core Module
Requirement ID	FR-02
Test Folder	tests/product
Page Object	pages/product.page.js
Test Case Count	15
Objective	Validate catalog navigation, product listing pages, product details, sorting controls, and product search.

3.2.3 Cart

Field	Details
Module Type	Core Module
Requirement ID	FR-03
Test Folder	tests/cart
Page Object	pages/cart.page.js
Test Case Count	15

Objective	Validate cart access, empty cart messaging, add-to-cart controls, checkout progress labels, and cart-related links.
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3.2.4 Wishlist and Compare

Field	Details
Module Type	Core Module
Requirement ID	FR-04
Test Folder	tests/wishlist-compare
Page Object	pages/wishlistCompare.page.js
Test Case Count	15
Objective	Validate wishlist, compare products, recently viewed products, email-a-friend entry points, and supporting links.

3.2.5 User Profile and Account

Field	Details
Module Type	Core Module
Requirement ID	FR-05
Test Folder	tests/user-profile
Page Object	pages/userProfile.page.js
Test Case Count	15
Objective	Validate account access rules for guest users, registration form fields, account footer links, and account recovery paths.

3.2.6 Address and Shipping

Field	Details
Module Type	Core Module

Requirement ID	FR-06
Test Folder	tests/address-shipping
Page Object	pages/addressShipping.page.js
Test Case Count	15
Objective	Validate shipping and returns content, address redirects, product availability text, and checkout address/shipping labels.

3.2.7 Add on - Checkout and Payment

Field	Details
Module Type	Add on Module
Requirement ID	FR-07
Test Folder	tests/add-on-checkout-payment
Page Object	pages/checkoutPayment.page.js
Test Case Count	15
Objective	Add coverage for checkout progress, empty-cart checkout behavior, gift card fields, add-to-cart availability, and price information.

3.2.8 Add on - Customer Support and Information

Field	Details
Module Type	Add on Module
Requirement ID	FR-08
Test Folder	tests/add-on-customer-support
Page Object	pages/customerSupport.page.js
Test Case Count	15
Objective	Add coverage for contact us, sitemap, privacy, conditions, about, news, blog, newsletter, social links, and footer information.

3.3 Browser Execution Matrix

Browser Project	Functional Test Cases	Execution Count
Chromium	120	120
Firefox	120	120
WebKit	120	120
Total	120 unique tests	360 browser executions

Milestone 4 - Reporting and CI/CD

4.1 Result Reporting Flow

The reporting flow is designed so that the trainer can review the project from GitHub without running the tests manually.

1. Project code is committed in the GitHub repository.
2. GitHub Actions starts the Playwright Allure workflow.
3. The configured browser projects execute the test suite.
4. Allure result data is generated from the Playwright execution.
5. The Allure HTML report is published to GitHub Pages.
6. The trainer opens the hosted report link and reviews execution status.

4.2 Report Links

Report Item	Link
GitHub Repository	https://github.com/AjayBhatnagar1712/Wipro_SDET_Playwright_Capstone
Hosted Allure Report	https://ajaybhatnagar1712.github.io/Wipro_SDET_Playwright_Capstone/

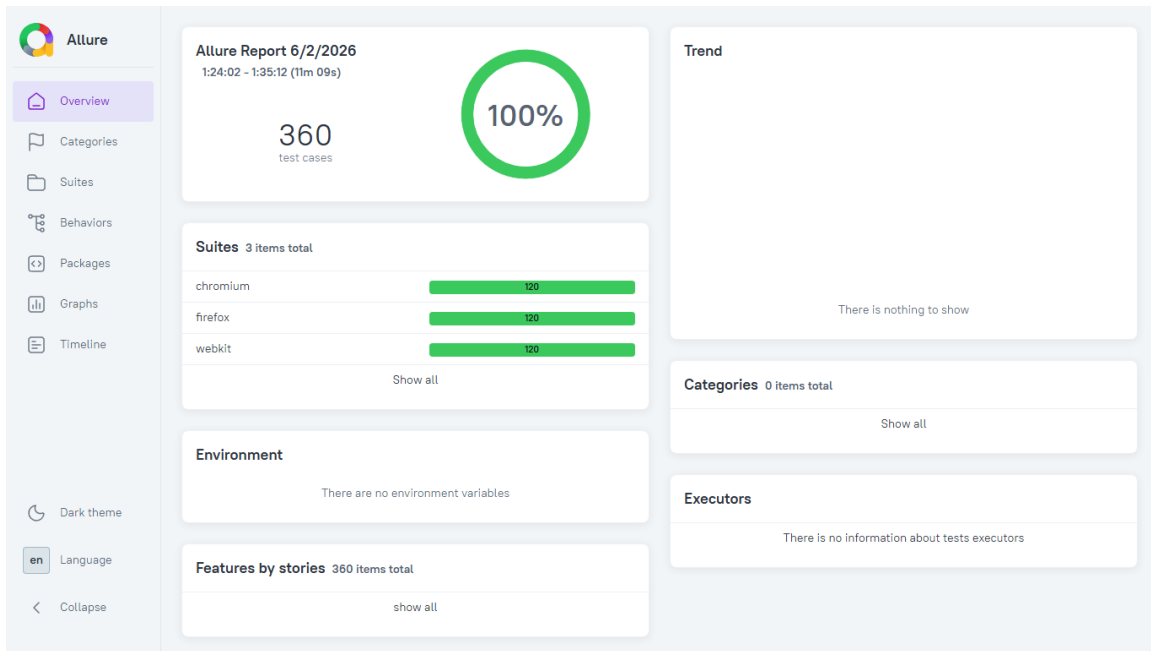


Figure 2: Hosted Allure report evidence from GitHub Pages.

4.3 Result Summary

Metric	Value
Core modules	6
Add on modules	2
Test cases per module	15
Total functional test cases	120
Configured browser projects	Chromium, Firefox, WebKit
Total browser executions	360
Reporting status	Allure report generated and hosted through GitHub Pages

Milestone 5 - Application Evidence

Screenshots are included as visual evidence for important pages covered by the automation suite.

Demo Web Shop
Sample Application by Tricentis

Register Log in Shopping cart(0) Wishlist(0)

Search store Search

BOOKS COMPUTERS ELECTRONICS APPAREL & SHOES DIGITAL DOWNLOADS JEWELRY GIFT CARDS

CATEGORIES

- Books
- Computers
- Electronics
- Apparel & Shoes
- Digital downloads
- Jewelry
- Gift Cards

MANUFACTURERS

- Tricentis

NEWSLETTER

Sign up for our newsletter:

Subscribe

Welcome, Please Sign In!

New Customer

By creating an account on our website you will be able to shop faster, be up to date on an orders status, and keep track of the orders you have previously made.

Register

Returning Customer

Email:

Password:

☐ Remember me?

Forgot password?

Log in

About login / registration

Put your login / registration information here. You can edit this in the admin site.

INFORMATION CUSTOMER SERVICE MY ACCOUNT FOLLOW US

Figure 3: Login page used for authentication checks.

Demo Web Shop
Sample Application by Tricentis

Register Log in Shopping cart(0) Wishlist(0)

Search store Search

BOOKS COMPUTERS ELECTRONICS APPAREL & SHOES DIGITAL DOWNLOADS JEWELRY GIFT CARDS

CATEGORIES

- Books
- Computers
- Electronics
- Apparel & Shoes
- Digital downloads
- Jewelry
- Gift Cards

MANUFACTURERS

- Tricentis

NEWSLETTER

Sign up for our newsletter:

Subscribe

Register

Your Personal Details

Gender: ☐ Male ☐ Female

First name: *

Last name: *

Email: *

Your Password

Password: *

Confirm password: *

Register

INFORMATION CUSTOMER SERVICE MY ACCOUNT FOLLOW US

Figure 4: Register page used for account creation field checks.

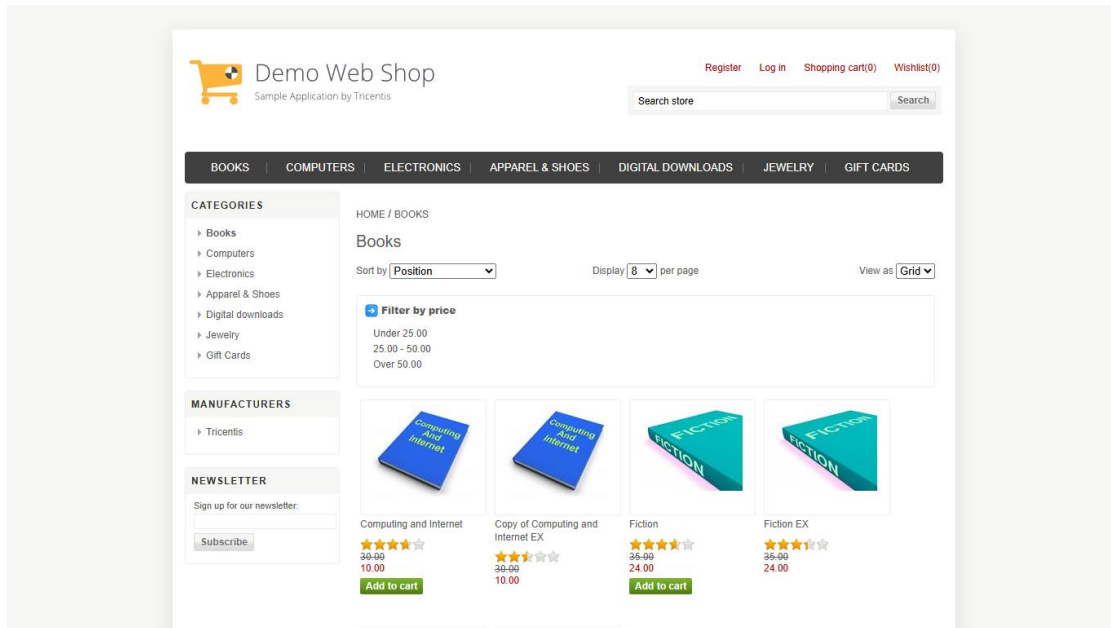


Figure 5: Books category page used for catalog checks.

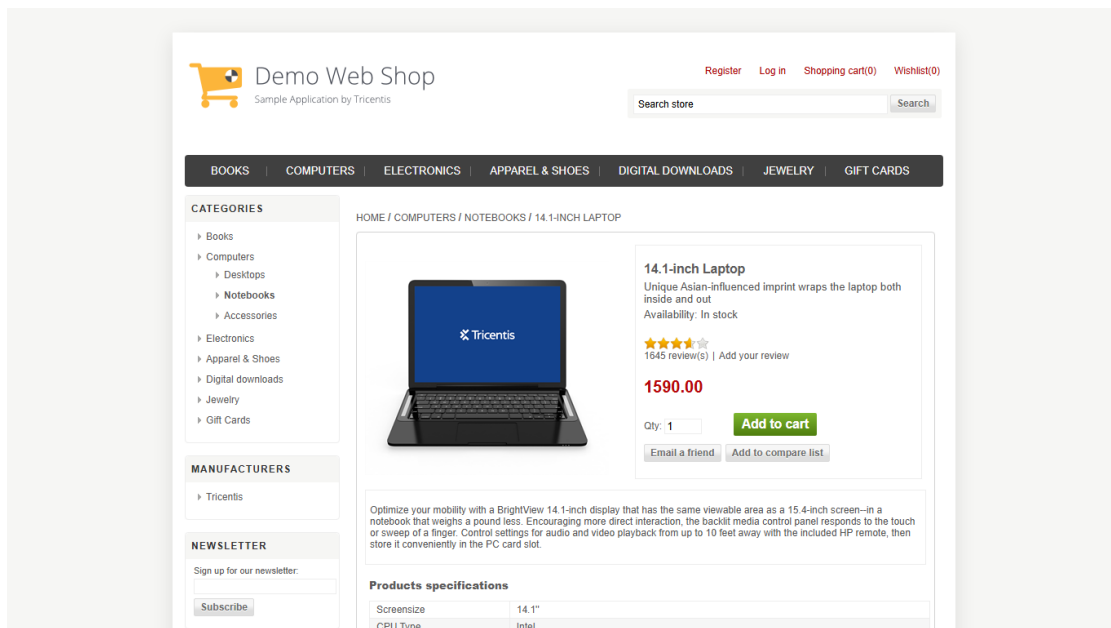


Figure 6: Product details page used for product, cart, and add on payment checks.

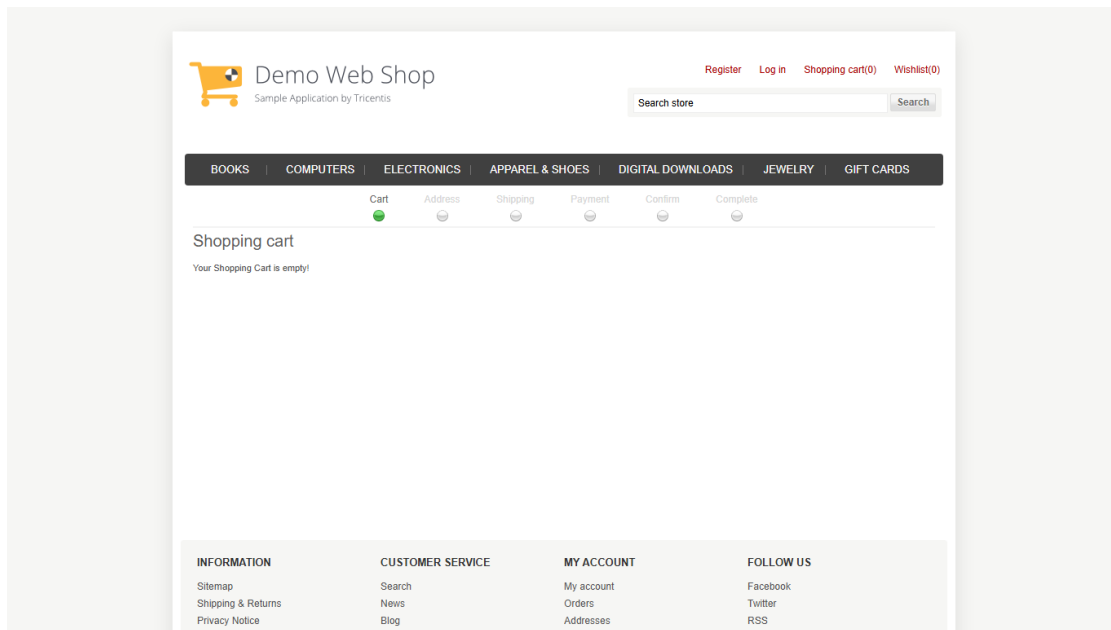


Figure 7: Empty cart page used for cart and checkout guard checks.

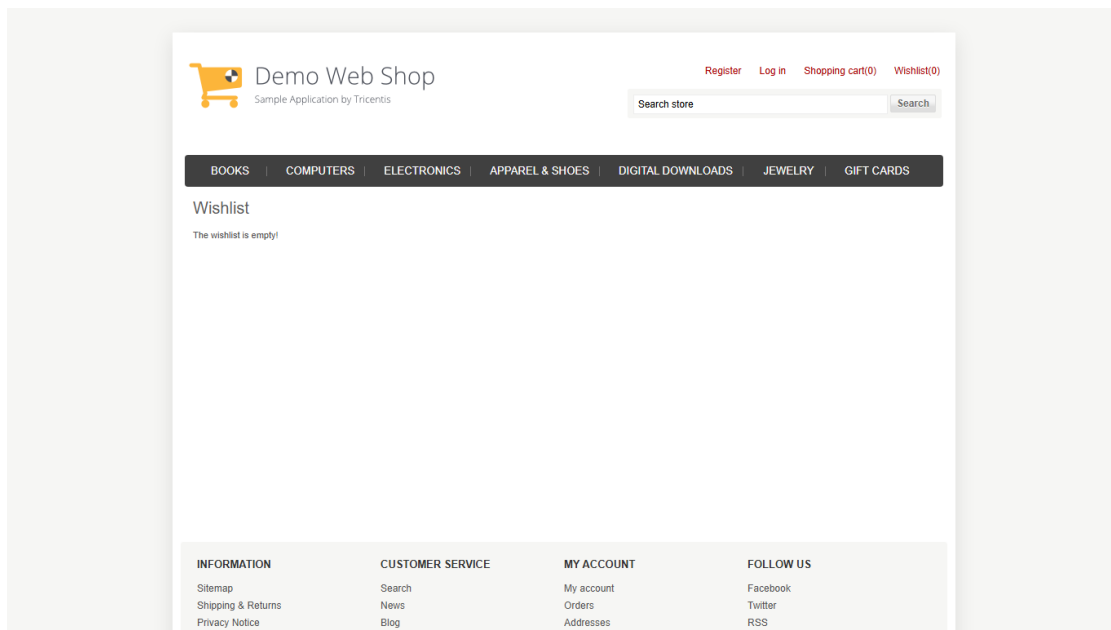


Figure 8: Wishlist page used for wishlist checks.

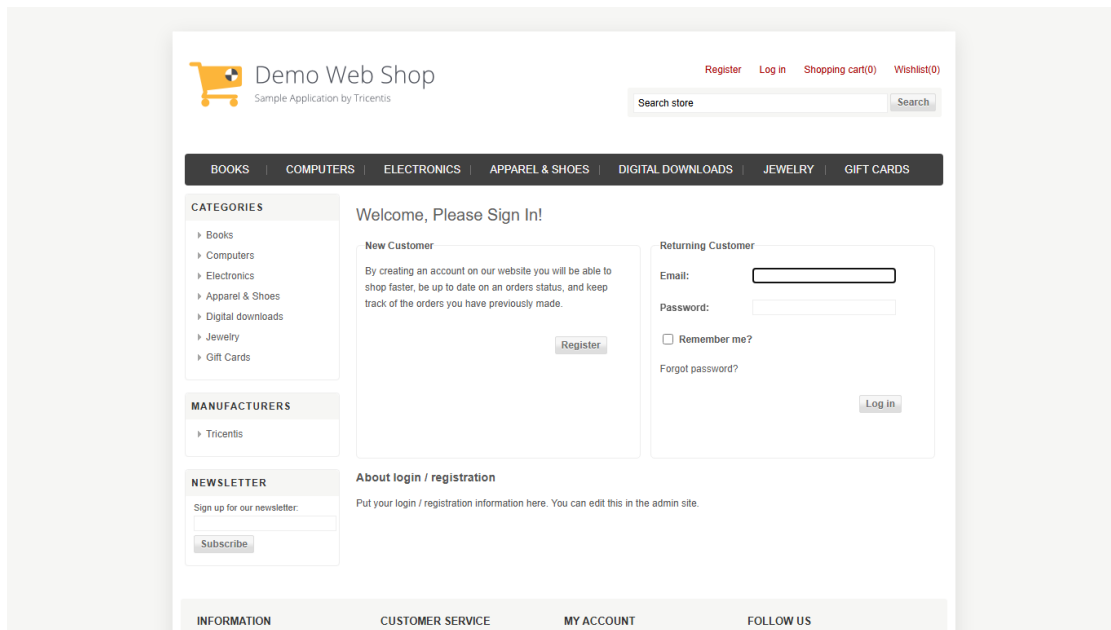


Figure 9: Protected account redirect evidence for guest users.

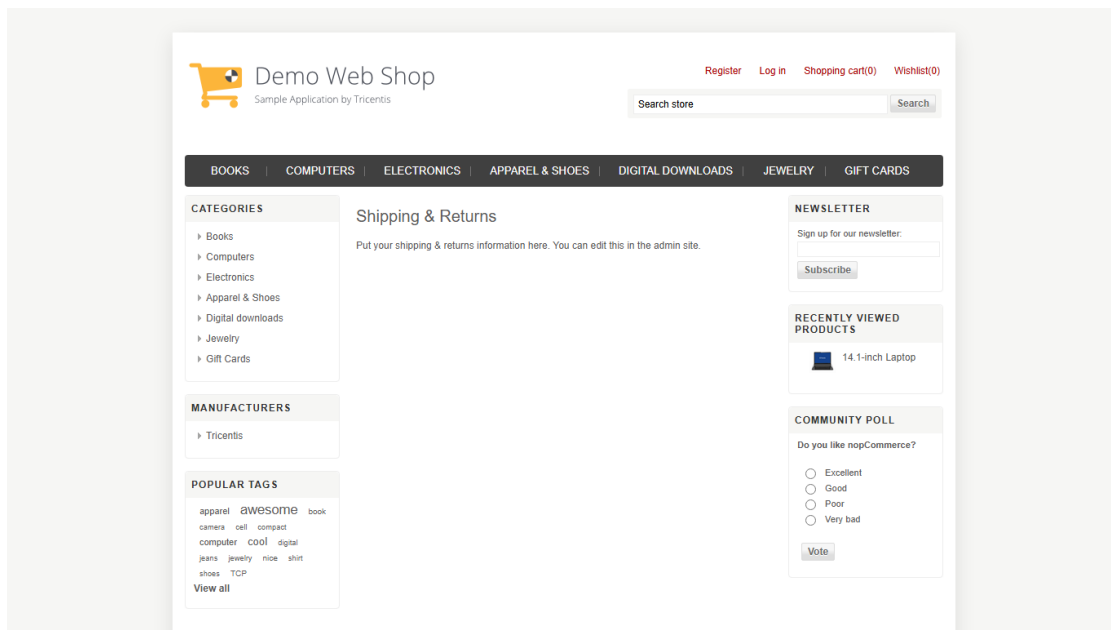


Figure 10: Shipping and returns page evidence.

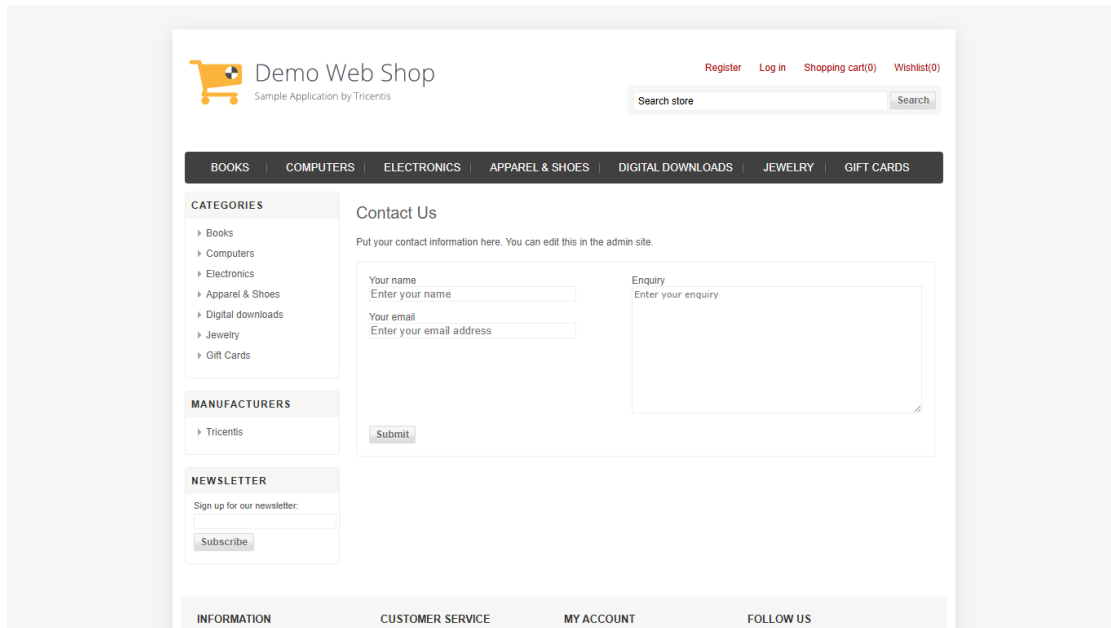


Figure 11: Contact us page evidence for support module.

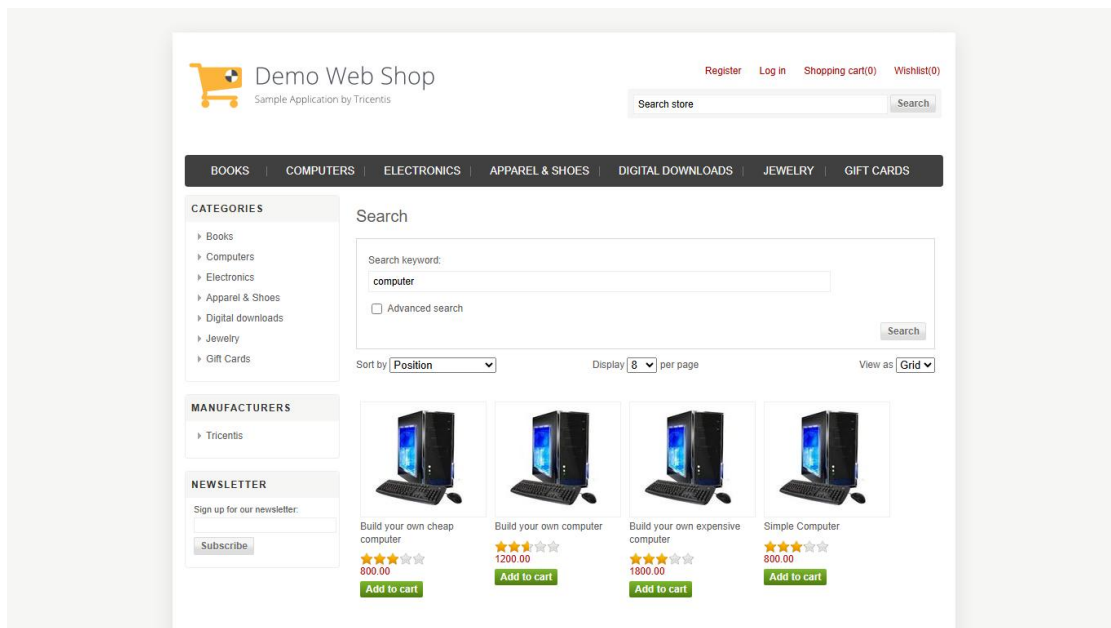


Figure 12: Search results page evidence.

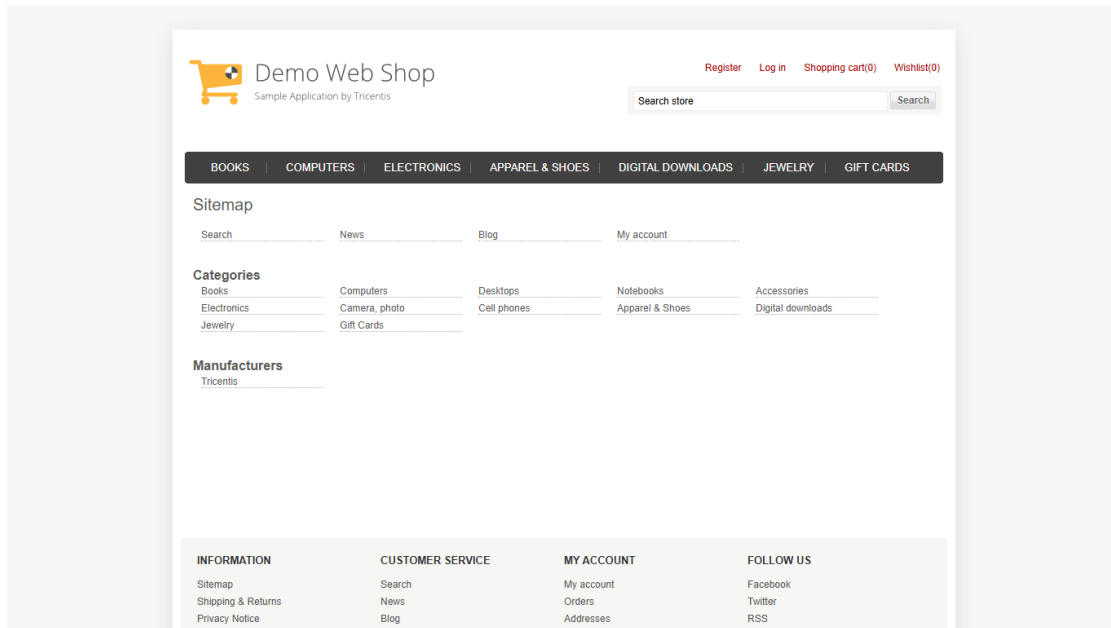


Figure 13: Sitemap page evidence for account, shipping, and information links.

Milestone 6 - Module Wise Test Cases

This section lists the final module-wise test coverage. Every module contains 15 functional test cases.

6.1 Authentication

Test Case ID	Validation Focus	Requirement
AUTH-001	Login page opens with correct title and heading	FR-01
AUTH-002	Login page shows email and password fields	FR-01
AUTH-003	Login page shows login button	FR-01
AUTH-004	Login page shows remember me and forgot password options	FR-01
AUTH-005	Login page shows registration option for new customer	FR-01
AUTH-006	Password recovery page opens correctly	FR-01
AUTH-007	Password recovery page has email field and recover button	FR-01
AUTH-008	Register page opens correctly	FR-01
AUTH-009	Register page shows gender choices	FR-01
AUTH-010	Register page shows personal detail fields	FR-01

AUTH-011	Register page shows password fields	FR-01
AUTH-012	Register button is available	FR-01
AUTH-013	Protected account page redirects guest user to login	FR-01
AUTH-014	Protected orders page redirects guest user to login	FR-01
AUTH-015	Header exposes login and register links	FR-01

6.2 Product and Search

Test Case ID	Validation Focus	Requirement
PROD-001	Home page shows welcome content and featured products	FR-02
PROD-002	Top navigation exposes catalog categories	FR-02
PROD-003	Books category shows product cards	FR-02
PROD-004	Computers category shows subcategories	FR-02
PROD-005	Electronics category shows subcategories	FR-02
PROD-006	Apparel and shoes category shows products	FR-02
PROD-007	Digital downloads category opens	FR-02
PROD-008	Jewelry category opens with product cards	FR-02
PROD-009	Gift cards category opens with products	FR-02
PROD-010	Build your own computer product details are visible	FR-02
PROD-011	Laptop product details are visible	FR-02
PROD-012	Smartphone product details are visible	FR-02
PROD-013	Book product page shows price and shipping text	FR-02
PROD-014	Category page has sort and page size controls	FR-02
PROD-015	Search page returns computer products	FR-02

6.3 Cart

Test Case ID	Validation Focus	Requirement
CART-001	Empty cart page opens	FR-03
CART-002	Empty cart message is shown	FR-03
CART-003	Header shopping cart link is visible	FR-03
CART-004	Cart footer link is visible	FR-03
CART-005	Book product has add to cart controls	FR-03
CART-006	Laptop product has add to cart controls	FR-03
CART-007	Computer product has add to cart controls	FR-03
CART-008	Gift card product has add to cart controls	FR-03
CART-009	Apparel product has cart controls	FR-03
CART-010	Jewelry product has cart controls	FR-03
CART-011	Cart page still shows search controls	FR-03
CART-012	Cart page shows order progress steps	FR-03
CART-013	Cart page shows information footer links	FR-03
CART-014	Cart page keeps footer follow links available	FR-03
CART-015	Cart page has checkout journey labels	FR-03

6.4 Wishlist and Compare

Test Case ID	Validation Focus	Requirement
WISH-001	Wishlist page opens	FR-04
WISH-002	Empty wishlist message is shown	FR-04
WISH-003	Header wishlist link is visible	FR-04
WISH-004	Gift card product has wishlist button	FR-04

WISH-005	Computer product has compare button	FR-04
WISH-006	Laptop product has compare button	FR-04
WISH-007	Compare products page opens	FR-04
WISH-008	Recently viewed products page opens	FR-04
WISH-009	New products page opens	FR-04
WISH-010	Product page exposes email a friend action	FR-04
WISH-011	Email a friend page opens	FR-04
WISH-012	Email a friend form has friend email field	FR-04
WISH-013	Email a friend form has message and send button	FR-04
WISH-014	Compare products footer link is available	FR-04
WISH-015	Wishlist footer link is available	FR-04

6.5 User Profile and Account

Test Case ID	Validation Focus	Requirement
USER-001	My account page requires login for guest	FR-05
USER-002	Order history page requires login for guest	FR-05
USER-003	Addresses page requires login for guest	FR-05
USER-004	Login page supports returning customer section	FR-05
USER-005	Footer exposes my account link	FR-05
USER-006	Footer exposes orders link	FR-05
USER-007	Footer exposes addresses link	FR-05
USER-008	Register page shows account creation form	FR-05
USER-009	Register page shows personal details fields	FR-05
USER-010	Register page shows email field	FR-05

USER-011	Register page shows password setup fields	FR-05
USER-012	Login page provides register button for new user	FR-05
USER-013	Login page provides forgot password link	FR-05
USER-014	Password recovery page supports account recovery	FR-05
USER-015	Sitemap includes account links	FR-05

6.6 Address and Shipping

Test Case ID	Validation Focus	Requirement
ADDR-001	Shipping returns page opens	FR-06
ADDR-002	Shipping returns page has customer service context	FR-06
ADDR-003	Guest addresses page redirects to login	FR-06
ADDR-004	Book product displays free shipping	FR-06
ADDR-005	Science product displays free shipping	FR-06
ADDR-006	Health book product displays delivery date	FR-06
ADDR-007	Laptop product shows in stock availability	FR-06
ADDR-008	Smartphone product shows in stock availability	FR-06
ADDR-009	Blue jeans product shows in stock availability	FR-06
ADDR-010	Shopping cart shows address step	FR-06
ADDR-011	Shopping cart shows shipping step	FR-06
ADDR-012	Footer has shipping returns link	FR-06
ADDR-013	Sitemap includes shipping returns link	FR-06
ADDR-014	Cart page has checkout progress section	FR-06
ADDR-015	Shipping page keeps information links available	FR-06

6.7 Add on - Checkout and Payment

Test Case ID	Validation Focus	Requirement
PAY-001	Empty cart keeps user on shopping cart page	FR-07
PAY-002	Empty cart blocks checkout with empty cart message	FR-07
PAY-003	Cart page is available before checkout	FR-07
PAY-004	Cart page shows payment step	FR-07
PAY-005	Cart page shows confirm step	FR-07
PAY-006	Cart page shows complete step	FR-07
PAY-007	Gift card product has recipient name field	FR-07
PAY-008	Gift card product has recipient email field	FR-07
PAY-009	Gift card product has sender fields	FR-07
PAY-010	Gift card product has message field	FR-07
PAY-011	Gift card product has add to cart button	FR-07
PAY-012	Gift card product shows fixed price	FR-07
PAY-013	Book product shows price information	FR-07
PAY-014	Laptop product shows price information	FR-07
PAY-015	Smartphone product shows price information	FR-07

6.8 Add on - Customer Support and Information

Test Case ID	Validation Focus	Requirement
INFO-001	Contact us page opens	FR-08
INFO-002	Contact us page has full name field	FR-08
INFO-003	Contact us page has email field	FR-08
INFO-004	Contact us page has enquiry textarea	FR-08

INFO-005	Contact us page has submit button	FR-08
INFO-006	Sitemap page opens	FR-08
INFO-007	Privacy notice page opens	FR-08
INFO-008	Conditions of use page opens	FR-08
INFO-009	About us page opens	FR-08
INFO-010	News page opens	FR-08
INFO-011	Blog page opens	FR-08
INFO-012	Footer information links are visible	FR-08
INFO-013	Footer customer service links are visible	FR-08
INFO-014	Newsletter subscription controls are visible	FR-08
INFO-015	Footer social links are visible	FR-08

Risks, Limitations, and Improvements

Area	Observation	Handling
Public website dependency	The website is not owned by the tester and can change in the future.	Assertions are written around stable visible content and common pages.
Real checkout	Actual payment processing cannot be completed on the demo site.	Checkout-related coverage is handled through cart guards, payment labels, gift card fields, and price checks.
Guest data	Creating permanent test users can make execution unstable.	The suite uses guest-user checks and form visibility validations.
Network behavior	Remote execution can be slower than local execution.	Timeouts, one worker, traces, screenshots, and video on failure are configured.
Future improvement	API and database validation can be added if a controlled backend is available.	The current project keeps UI automation stable for the selected public website.

Conclusion

The Demo Web Shop Playwright capstone project satisfies the updated project requirement by organizing the suite into 6 core modules and 2 add on modules. Each module has 15 functional test cases, giving 120 functional test cases in total.

The same tests are configured for Chromium, Firefox, and WebKit, giving 360 browser executions when the full cross-browser suite is run. The project also demonstrates Page Object Model usage, reusable data, Allure reporting, GitHub Actions CI/CD, and GitHub Pages hosting.

GitHub repository: https://github.com/AjayBhatnagar1712/Wipro_SDET_Playwrite_Capstone

Hosted Allure report: https://ajaybhatnagar1712.github.io/Wipro_SDET_Playwrite_Capstone/